

What English-tuned ablation misses in Slovene

Framing

Abliteration removes refusal behaviour from language models by orthogonalising weight matrices against a "refusal direction" extracted from English harmful-vs-harmless contrasts [1]. The procedure was introduced by Ardit et al. [1] and automated by tools such as Heretic, which uses TPE search over layer, position, direction index, and LoRA rank to find ablation configurations that minimise an English refusal objective while bounding KL divergence from the original model. The question motivating this study is whether that English-centred objective leaves Slovene refusal behaviour uncontrolled. If the English refusal direction is universal across languages, as Wang et al. [2] report for safety-aligned models evaluated across 14 languages, then ablation tuned on English should transfer to Slovene. If instead Slovene refusal relies partly on representations that the English objective does not observe, a gap will remain. The practical consequence is that a model ablated to comply in English might still refuse in Slovene, or might lose Slovene capabilities that the English-only KL constraint does not protect.

Several concurrent lines of work inform this question. BabelSteering [4] and Yoon et al. [5] show that safety alignment imposes disparate costs across languages, with non-English users bearing higher utility loss. Aziz et al. [6] diagnose the breakdown point for low-resource safety as an action failure rather than a representation failure: the model encodes harmfulness correctly but fails to convert the representation into refusal. Wu et al. [7] formalise this as a disentangled safety hypothesis, separating a recognition axis from an execution axis. On the ablation side, Fafula [8] documents off-target effects of refusal removal across model families, and Young [14] and Petrov [15] compare ablation methods across architectures. Upadhyaya and Sikdar [9] analyse how safety and language identity become entangled, while Hawkins et al. [10] show that benign multilingual fine-tuning has heterogeneous safety impacts. Labunets [11] demonstrates that refusal geometry reflects refusal training diversity. On quantisation effects relevant to our NF4 setup, Marchisio et al. [12] and Chimoto et al. [13] show that quantisation calibration language matters for multilingual performance. Tang et al. [16] identify language-specific neurons, and Ghussin et al. [17] propose principled layer selection for multilingual steering. Frank [18] argues that refusal-based alignment evaluation conflates detection with routing.

We study this question in a controlled bilingual setting using two sibling 12B models from the Gemma-3 family: google/gemma-3-12b-it (the English-centric reference) and cjvt/GaMS3-12B-Instruct (a Slovene continual-pretraining and SFT variant of the same architecture). Both are loaded in bnb_4bit NF4 quantisation throughout. The study proceeds in two iterations. Iteration 1 runs matched Heretic ablation on both models, screens for cross-language refusal-direction transfer, and builds the shared bilingual evaluation dataset. Iteration 2 uses that dataset to measure the surrogacy gap (what English outcomes fail to predict about Slovene), test causal mechanisms, and evaluate safety and utility on held-out benchmarks.

[Figure fig_behaviour_summary: generated in the report phase, not in this draft]

Iteration 1

Experiment 1: Matched Heretic runs

Goal. Run identical Heretic TPE searches on both models under the same seed, select an ablated checkpoint for each, and compare the two models' responses to the same edits.

Setup. Both models were searched over 116 complete trials (seed 20260923) using Heretic (SHA 3521f864) with the default configuration: `orthogonalize_direction = true`, `row_normalization = 'full'`, rank-3 LoRA. The English keyword refusal rate (out of 100 harmful prompts) and mean KL divergence served as Heretic's two objectives. Selection followed a frozen protocol: minimise KL subject to refusals ≤ 10 ; if no trial met this primary criterion, fall back to minimising refusals subject to KL ≤ 1.0 . Both models fell back to the secondary rule.

Selected trials. GaMS3 trial 88 reached 16/100 English refusals at KL 0.175. Gemma trial 96 reached 69/100 English refusals at KL 0.024. The Gemma model refused far more often than GaMS3 under identical ablation parameters, and no Gemma trial achieved $\leq 50/100$ refusals. The ratio of median refusal counts (Gemma / GaMS3) across the 60 shared startup edits was $13.4 \times [6.9, 30.4]$. In all 60 startup edits, Gemma refused at least as often as GaMS3 (60/60, median paired difference +25).

Behaviour on the selected trials. The table below shows keyword refusal rates (out of 100 harmful prompts) and FLORES+ NLL for each checkpoint.

Model	Checkpoint	EN refusals	SL refusals	KL mean	FLORES NLL	EN FLORES NLL SL
GaMS3	original	98	97	0.000	3.332	2.570
GaMS3	own edit (trial 88)	16	10	0.175	3.334	2.566
GaMS3	swap (Gemma trial 96)	25	25	0.046	3.336	2.576
Gemma	original	100	100	0.000	3.930	3.427
Gemma	own edit (trial 96)	69	85	0.024	3.929	3.432
Gemma	swap (GaMS3 trial 88)	91	95	0.293	3.944	3.455

GaMS3's own edit reduced English refusals from 98 to 16 and Slovene refusals from 97 to 10. Slovene refusal dropped even though Heretic optimised only on English, and FLORES NLL changed by at most 0.01. For Gemma, the edit reduced English refusals from 100 to 69 but Slovene refusals only from 100 to 85. Swapping edits between models did not reproduce the other model's behaviour: GaMS3 under Gemma's edit showed 25/25 (EN/SL) refusals, and Gemma under GaMS3's edit showed 91/95, both far from the original model's own-edit performance.

Direction geometry. Across layers 24--47, the mean cosine between the two models' diff-in-means refusal directions was 0.57; across all layers it was 0.63. The cosine between their harmless-token directions was 0.98. Random directions gave a mean $|\cos|$ of 0.014. The refusal directions are more aligned than chance but far from parallel; the harmless directions are near-identical, consistent with the shared tokenizer and architecture.

A3 screen (paired startup edits). The Spearman correlation of refusal counts between the two models across 60 paired edits was 0.78 [0.63, 0.88]. KL divergence correlated at 0.97. However, the incremental variance explained by the sibling model's outcome (`dR2_sibling`) was near zero for both directions of prediction, meaning the two models' refusal counts are correlated through shared parameters but one model's outcome does not add predictive value over those parameters. The screen verdict was "shared via strength only": the correlation is driven by parameter strength, not by a direct refusal-mechanism coupling.

Sanity checks. All 60 startup edits used identical parameter draws (verified). FLORES NLL shifts

were ≤ 0.01 for both models. Slovene marker validation on the selected GaMS3 edit showed accuracy 0.75 and kappa 0.48 (fair agreement between keyword markers and LLM-judge labels).

Dataset construction

Goal. Build a frozen bilingual (EN/SL) evaluation dataset that spans safety, utility, and mechanistic probing, using matched translation quality controls.

Result. The dataset contains 48,696 rows across 10 blocks (S1--S7 families): S1 (Heretic direction/KL calibration, 2000 rows), S2 (semantic harmful/harmless pairs, 1664), S3 screen prompts (540 prompts + 640 utility items), S4 (StrongReject harmful/harmless twins, 1028), S5 (RefusEU [3], 2800), S5X (RefusEU cross-translations, 1400), S6 (XSTest over-refusal, 900), S7 (Slovenian LLM eval tasks including ARC, BoolQ, HellaSwag, 35,700; FLORES + devtest, 2024). Protocol hash: dc33bde4.

Translation. Slovene translations were produced by Gemini-2.5-flash (temperature 0) as the primary translator, with NLLB-200-distilled-1.3B as fallback for 8 rows where Gemini refused. Mean back-translation chrF was 81.25; 3 rows fell below chrF 40. Mean LaBSE cross-lingual similarity was 0.856. Quality control was automated only (back-translation chrF, LaBSE, GlotLID); no native-speaker review was conducted.

Splits. Items are assigned to halves A and B by SHA-1 hash of their semantic ID. This split is used throughout iteration 2 for placebo-controlled surrogacy analyses: English traits computed on half A predict English (placebo) and Slovene (test) on half B.

Experiment 3: A1 screen, EN vs. SL refusal-direction transfer

Goal. Screen whether the English-extracted refusal direction transfers to Slovene, using a 2×2 matrix (source language $\times$ evaluation language), increment tests, and multiple controls. This was a screen, not a confirmatory test.

Setup. For each model, the experiment extracted refusal directions from half-A items using diff-in-means on harmful vs. harmless activations, separately for English and Slovene. It then applied 15 ablation conditions per model (activation steering at inference) to half-B items and measured refusal rates using keyword markers and an LLM judge. The judge was gpt-4.1 for 1,596 outputs (recovered from logs after an API budget exhaustion) and Qwen3-14B locally for the remainder.

Precision deviations. Several deviations from the original plan were recorded. The Gemma candidate grid was cut to every-second layer and positions $\{-1, -5\}$ for GPU time constraints. Translation used a system/user wrapper prompt because bare Gemini prompts elicited refusal. NLLB served as fallback for 22/1070 translation rows. Winsorization ($q = 0.995$) was applied to directions because one massive-activation dimension dominated the raw diff-in-means.

Validity gates. The R validity gate (refusal readout Spearman correlation with judged refusal across 16 conditions) passed for Gemma (Spearman 0.91, AUROC 0.96) but failed for GaMS3 (Spearman 0.51, AUROC 0.70). The GaMS3 failure means the keyword-based refusal readout did not track actual refusal behaviour well enough for the screen's quantitative tests to be trusted. Per the frozen protocol (rule F6), the screen's primary outcome falls to the judged refusal rate for GaMS3.

GaMS3 results. Because the R validity gate failed, quantitative screen results for GaMS3 are exploratory. The baseline English refusal score was 5.1 (arbitrary units); Slovene was 2.0. The 15 ablation conditions showed varied effects, but no reliable conclusion about EN → SL transfer could be drawn from the GaMS3 data.

Gemma results. The Gemma validity gate passed. The key result: applying the English refusal direction (d_EN) at inference removed most English refusal (from 99% to low values depending on the condition), but Slovene refusal survived at 0.85 residual on a matched-efficacy comparison. The predicted EN → SL transfer was reversed: the screen expected the English direction to reduce Slovene refusal proportionally, but instead SL refusal persisted.

Screen verdict. WEAK. The predicted cross-language transfer contrast was reversed for Gemma. For GaMS3, the validity gate failure prevented a quantitative verdict. The screen provided the initial signal that English ablation may miss Slovene refusal, motivating the confirmatory analyses in iteration 2.

Matched-efficacy grid. Conditions rescaled so that English refusal dropped by equal amounts showed that Slovene refusal systematically persisted more than English refusal at every matched English decrement. The transfer matrix (source × evaluation language) confirmed that English directions removed English refusal but left substantial Slovene refusal intact.

Addition dose-response. Increasing the steering coefficient (scaling up the English refusal direction) reduced both English and Slovene refusal, but Slovene always lagged behind English: at every dose level, Slovene retained more refusal than English.

Iteration 2

Iteration 2 built on the screen results from iteration 1 by running five pre-registered experiments that measured the surrogacy gap, tested causal mechanisms, and evaluated safety and utility on held-out data.

Experiment 5: Utility and mechanism

Goal. Evaluate whether the selected ablation edits preserve utility across both languages and characterise the internal geometry of the harm representation before and after editing.

T1: Harmful behaviour (S4 harmful items, 257 per cell)

The table below shows judged refusal rates on held-out harmful items (S4, 257 items per model × checkpoint × language). The gpt-4.1 judge labelled a random ~52% subsample; remaining items used a second-family free judge (Nemotron-3-ultra-550b); markers show the Heretic keyword rule on all 257.

Model	Ckpt	Lang	Refused % [95% CI]	Complied % [95% CI]
GaMS3	orig	EN	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
GaMS3	orig	SL	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
GaMS3	edit	EN	5.6 [2.4, 9.5]	92.9 [88.1, 96.8]
GaMS3	edit	SL	6.7 [3.0, 11.1]	89.6 [84.4, 94.1]
Gemma	orig	EN	99.3 [97.8, 100.0]	0.7 [0.0, 2.2]
Gemma	orig	SL	99.2 [97.6, 100.0]	0.8 [0.0, 2.4]

Gemma edit	EN	70.3 [61.9, 78.0]	24.6 [16.9, 32.2]
Gemma edit	SL	95.3 [92.0, 98.7]	4.0 [1.3, 7.3]

For GaMS3, the edit reduced refusal from 100% to ~6% in both languages, with no meaningful EN-SL gap. For Gemma, the edit reduced English refusal from 99% to 70% but left Slovene refusal at 95%. This 25-percentage-point gap (70.3% vs. 95.3%) confirms the pattern seen in the iteration-1 screen: the English-tuned Heretic edit substantially missed Slovene refusal in Gemma.

Judge agreement: second judge vs. gpt-4.1 on a stratified sample of 200 gave 6-way kappa 0.765, refused-vs-not kappa 0.828, raw agreement 0.885.

T2: KL divergence on harmless twins (S4, 128-token generations)

Model	Ckpt	Lang	KL1 median [95% CI]	KL32 median [95% CI]	Benign refusal %	Malformed %	Empty %
GaMS3	orig	EN	0 (ref)	0 (ref)	0.0	0.0	0.0
GaMS3	orig	SL	0 (ref)	0 (ref)	0.0	0.0	0.0
GaMS3	edit	EN	0.0441 [0.0323, 0.0530]	0.0228 [0.0196, 0.0250]	0.0	0.0	0.0
GaMS3	edit	SL	0.0174 [0.0152, 0.0211]	0.0068 [0.0061, 0.0086]	0.0	0.0	0.0
Gemma	orig	EN	0 (ref)	0 (ref)	0.0	0.0	0.0
Gemma	orig	SL	0 (ref)	0 (ref)	9.5	0.4	0.0
Gemma	edit	EN	0.0000 [0.0000, 0.0000]	0.0150 [0.0125, 0.0171]	0.0	0.0	0.0
Gemma	edit	SL	0.0113 [0.0083, 0.0163]	0.0037 [0.0030, 0.0046]	0.0	0.4	0.0

KL1 values ranged from 0.011 to 0.044. Gemma EN edit showed KL1 median 0.0000 (at the first token the edit barely changed harmless outputs). Output validity was clean across all cells.

T3: Utility (0-shot, 250 items per task per language)

Model	Lang	Macro original	Macro edited	Change [95% CI]	Holm p
GaMS3	EN	72.5	72.7	+0.13 [-0.53, +0.80]	1.000
GaMS3	SL	64.9	65.1	+0.20 [-0.73, +1.13]	1.000
Gemma	EN	71.5	71.6	+0.13 [-0.53, +0.80]	1.000
Gemma	SL	62.1	62.1	+0.00 [-0.60, +0.60]	1.000

All macro changes were ≤ 0.20 points. All Holm-corrected p-values were 1.000 over the 4 macro comparisons. Individual task changes ranged from -1.20 to +1.60 points; none reached significance. The edits did not measurably alter utility in either language for either model.

T4: FLORES+ per-token NLL

Model	Lang	NLL original	NLL edited	Change [95% CI]
GaMS3	EN	3.3072	3.3068	-0.0004 [-0.0025, +0.0016]
GaMS3	SL	2.4524	2.4526	+0.0002 [-0.0012, +0.0016]
Gemma	EN	3.9194	3.9174	-0.0020 [-0.0031, -0.0008]
Gemma	SL	3.4289	3.4293	+0.0004 [-0.0007, +0.0016]

NLL shifts were ≤ 0.002 for all cells. The Gemma English NLL decrease (-0.002) was statistically

below zero but negligible in magnitude.

T5: Bilingual harm geometry (original models, primary site)

Model	Layer	cos(d_EN, d_SL)	Corrected CV AUROC	DiM EN/SL	S4 frozen DiM EN/SL	cos(lang dir, d_EN)	Shuffled AUROC
GaMS3	34	0.832	0.890	0.963 / 0.945	0.998 / 0.997	0.035	0.490
Gemma	20	0.919	0.979	0.979 / 0.941	0.997 / 0.996	0.109	0.515

Cross-validation AUROC on S3 development items exceeded 0.94 for both models in both languages. Cross-language transfer on held-out S4 items exceeded 0.995 in both directions for both models. The harm direction is shared across languages. The language-identity direction is near-orthogonal to d_EN (cos 0.035 and 0.109). Shuffled-label AUROC was at chance (0.49--0.52), confirming the directions are not artifacts of data leakage.

Both models encode harmfulness similarly across EN and SL in their representations. The harm direction is shared; the language direction is separate. For Gemma, the behavioural gap (70.3% vs. 95.3% residual refusal) cannot be attributed to a representational difference: the model knows the prompt is harmful in both languages, but acts on that knowledge differently.

T6: Frozen vs. refit probes (S4, primary site)

Model	Lang	Probe	Frozen orig AUROC	Frozen edit-orig [95% CI]	Refit edit-orig [95% CI]	Class
GaMS3	EN	dim	0.998	-0.401 [-0.434, -0.368]	-0.014 [-0.021, -0.007]	INFO-LOSS
GaMS3	EN	lr	0.999	-0.596 [-0.623, -0.570]	-0.000 [-0.001, 0.001]	DRIFT
GaMS3	SL	dim	0.997	-0.576 [-0.611, -0.542]	-0.043 [-0.056, -0.030]	INFO-LOSS
GaMS3	SL	lr	0.996	-0.661 [-0.691, -0.635]	-0.000 [-0.002, 0.002]	DRIFT
Gemma	EN	dim	0.997	-0.000 [-0.001, 0.000]	-0.000 [-0.001, 0.000]	PRESERVED
Gemma	EN	lr	0.996	-0.001 [-0.002, 0.000]	-0.000 [-0.000, 0.000]	PRESERVED
Gemma	SL	dim	0.996	-0.000 [-0.001, 0.000]	-0.000 [-0.001, 0.000]	PRESERVED
Gemma	SL	lr	0.989	-0.002 [-0.003, -0.001]	0.000 [-0.000, 0.000]	DRIFT

All-layer profile (minimum frozen-probe AUROC on edited activations):

Model	Lang	Min frozen-edit AUROC (layer)	Refit AUROC there
GaMS3	EN	0.201 (L27)	0.993
GaMS3	SL	0.196 (L27)	0.987
Gemma	EN	0.507 (L4)	0.709
Gemma	SL	0.492 (L6)	0.740

For GaMS3, the frozen probe collapsed after editing: AUROC dropped to 0.201 at the worst layer, but a refit probe recovered to 0.993. The edit moved the harm signal away from its original subspace (INFO-LOSS on dim, DRIFT on logistic), but the information persisted in a different location.

For Gemma, the frozen probe was preserved: AUROC dropped by at most 0.002. The edit did not move the harm representation.

The contrast is informative. GaMS3's edit reached low refusal in both languages by destroying the harm signal at the probe layer. Gemma's edit left the harm signal intact, consistent with its failure to suppress Slovene refusal: the model still encodes harmfulness, but the readout mechanism was

partially disrupted for English while the Slovene readout remained functional.

T7: Drift geometry

For GaMS3, Cohen's d on the frozen harm axis dropped from 6.20 to 0.30 (EN) and from 5.61 to -0.18 (SL). The share of mean-difference energy on the frozen axis dropped from 0.71 to 0.002 (EN). The cosine between the pre- and post-edit refit axes was 0.274 (EN) and 0.188 (SL), confirming the edit rotated the harm axis. Collapse onset was at layer 22.

For Gemma, Cohen's d dropped from 5.29 to 4.94 (EN) and from 3.76 to 3.50 (SL). The frozen axis retained most of its energy share (0.83 → 0.80). The cosine between refit axes was 0.981. The edit barely moved the harm signal.

T8: Readout validity gate (S4 judged items)

Model	Lang	R_seq item AUROC	R_seq 16-cell Spearman	R_seq pass	R1 item AUROC	R1 pass	Primary trait
GaMS3	EN	0.989	0.867	True	0.979	False	R_seq
GaMS3	SL	0.995	0.899	True	0.963	False	R_seq
Gemma	EN	0.804	0.861	False	0.897	False	JUDGE_ONLY
Gemma	SL	0.920	0.856	True	0.968	True	R_seq

GaMS3 R_seq passed in both languages. Gemma R_seq passed only for SL; EN was classified as JUDGE_ONLY, meaning the judge labels should be trusted over the R_seq score for Gemma EN cells. R1 did not pass for GaMS3 in either language.

T9: Criterion shift vs. evidence loss

Both models showed EVIDENCE_LOSS under both R_seq and R1 readouts: the slope of the decision function decreased after editing, indicating the internal signal weakened rather than the threshold shifting. For GaMS3, the R_seq slope ratio was 0.36 (EN) and 0.42 (SL). For Gemma, the R_seq slope ratio was 0.61 (EN) and 0.48 (SL).

T10: r_prior geometry (original models)

The false-refusal direction r_prior (constructed from prompts judged harmless that the model refuses, orthogonalised to d_EN) was defined for both models. For GaMS3, 36 refusals (15 EN, 21 SL) out of 275 r_prior generations; cos(r_prior, d_EN) before orthogonalisation was 0.824 (pooled). For Gemma, 55 refusals (8 EN, 47 SL) out of 273; cos(r_prior, d_EN) was 0.631. The within-SL cos(r_prior, lang_dir) was -0.038 (GaMS3) and -0.139 (Gemma), confirming r_prior is not simply the language direction.

T12: Edit-strength dose-response (exploratory)

Model	Scale f	Marker refusal EN %	Marker refusal SL %	KL1 median EN	KL1 median SL
GaMS3	0.0	95.8	100.0	0.0000	0.0000
GaMS3	0.5	75.0	68.8	0.0088	0.0037
GaMS3	1.0	0.0	0.0	0.0405	0.0177
GaMS3	1.5	0.0	2.1	0.1610	0.1167
Gemma	0.0	93.8	100.0	0.0000	0.0000
Gemma	0.5	60.4	95.8	0.0000	0.0046
Gemma	1.0	27.1	91.7	0.0000	0.0103

Gemma 1.5	22.9	68.8	0.0000	0.0303
Gemma 2.0	12.5	31.2	0.0000	0.0924
Gemma 3.0	4.2	8.3	0.0000	0.8228

For GaMS3, scaling the LoRA edit by $f = 0.5$ brought both languages to 68--75% refusal; at $f = 1.0$ both reached 0%. EN and SL refusal declined in parallel. For Gemma, at $f = 1.0$ (the core edit) EN refusal was 27% while SL was 92%. Reaching SL refusal below 10% required $f = 3.0$, at which point KL divergence rose to 0.82 (SL), well above Heretic's operational range. This dose-response confirms that the EN-SL gap in Gemma is not an artifact of edit strength: at every dose level, SL refusal exceeded EN refusal.

[Figure fig_dose_response: generated in the report phase, not in this draft]

T13: GaMS3 base model diagnostic (pre-instruction, descriptive)

The uninstructed cjvt/GaMS3-12B base model (before SFT) was tested with a Question/Answer format at layer 34. Frozen S3→S4 diff-in-means AUROC was 0.817 (EN) and 0.773 (SL). Grouped-CV S4 AUROC reached 0.988 (EN) and 0.958 (SL). The cosine between the base model's EN and SL harm directions was 0.781; the cosine between the base model's d_{EN} and the instruction-tuned model's d_{EN} was 0.305. Out of 20 EN harmful continuations, 14 complied and 5 refused; out of 20 SL harmful continuations, 13 complied and 5 refused. The base model already encodes a harm axis that partially survives instruction tuning, but the low cosine with the instruct model's direction (0.305) indicates instruction tuning rotated the harm representation.

Experiment 4: C1 behaviour, RefusEU evaluation

Goal. Evaluate all checkpoints (originals, edits, community ablation reference) on the held-out RefusEU multilingual refusal benchmark [3] in both EN and SL, measuring refusal rate, attack success rate, over-refusal, and response validity.

Judge. The primary judge was a local Qwen3-14B model (agreement with gpt-4.1: kappa 0.83 on 6-way labels, 0.91 on refused-vs-not; tested on a calibration set before deployment). The gpt-4.1 judge was blocked partway through by API budget limits, so local Qwen3-14B served as the sole judge for all 4,800 generations.

Headline RefusEU (S5) refusal rates:

Model	Checkpoint	EN S5 refusal	SL S5 refusal	EN S6 over-refusal	SL S6 over-refusal
GaMS3	original	0.986	0.871	0.100	0.107
GaMS3	edit	0.014	0.000	0.000	0.000
Gemma	original	0.971	0.939	0.087	0.327
Gemma	edit	0.287	0.739	0.013	0.193
Community ref -		0.054	0.114	0.007	0.027

All changes from original to edit were significant (Holm-corrected $p < 1e-14$ for refusal reductions). Invalid rate was 0.000 across all cells.

GaMS3. The edit reduced EN refusal from 0.986 to 0.014 and SL refusal from 0.871 to 0.000. Both languages reached near-zero refusal. Over-refusal dropped from ~0.10 to 0.000 in both languages. GaMS3's baseline SL refusal was already lower than its EN refusal (0.871 vs. 0.986), and the edit

eliminated refusal in both languages.

Gemma. The edit reduced EN refusal from 0.971 to 0.287 and SL refusal from 0.939 to 0.739. The residual Slovene refusal (0.739) is $2.6 \times$ the residual English refusal (0.287). This is the headline result of the study: an English-tuned ablation edit left most Slovene refusal intact in Gemma while removing most English refusal. Over-refusal dropped in both languages but remained substantial in SL (0.193 vs. 0.013 in EN for the edit). The original Gemma model already had high SL over-refusal (0.327), and the edit reduced it only to 0.193.

Community reference ablation. A community Heretic ablation applied to Gemma reached 0.054 EN refusal and 0.114 SL refusal. This more aggressive edit removed most refusal in both languages but had higher FLORES NLL shifts (0.054 EN, 0.111 SL) than the selected trial.

Paired cross-language analysis (S5X, 100 matched pairs). For Gemma edit, the paired EN-SL refusal gap was $+0.69$ [0.60, 0.78]: on the same semantic content, Slovene refusal exceeded English refusal by 69 percentage points. The difference-in-differences (edit gap minus original gap) was $+0.66$ [0.55, 0.76]. For the community reference, the gap was only $+0.12$ [0.05, 0.20].

Experiment 6: GaMS3 surrogacy panel

Goal. Measure the English surrogacy gap for GaMS3: for each ablation trait, how much does an English outcome fail to predict the corresponding Slovene outcome?

Method. 246 Heretic edits (initial startup edits E0 plus TPE-selected edits E_TPE) were generated, activations collected on half-B items, and six traits measured: R (refusal, R_seq readout), R1 (R at first token), Rb (binary refusal), K (KL divergence), N (NLL), M (maximum logit). The primary learner was gradient-boosted trees. The surrogacy gap is defined as $\text{Gap}_t = R2(EN_A \rightarrow EN_B) - R2(EN_A \rightarrow SL_B)$, where $R2^*$ is noise-ceiling normalised. The blind share B_t estimates the systematic (edit-determined) portion of the unseen Slovene variation. 160 non-collapsed edits were fitted.

Gap table:

Trait	$R2^*$ placebo	$R2^*$ test	Gap	95% CI	Holm p	B_t [95% CI]
R	0.983	0.968	0.015	[0.006, 0.031]	NA	0.004 [-0.004, 0.009]
R1	0.998	0.956	0.042	[0.027, 0.070]	NA	0.015 [0.002, 0.030]
Rb	0.982	0.954	0.028	[0.012, 0.068]	NA	0.005 [-0.007, 0.013]
K	0.977	0.830	0.147	[0.074, 0.265]	0.000	0.050 [0.001, 0.121]
N	-0.225	0.417	-0.642	[-14.8, 0.98]	0.954	-
M	0.272	0.232	0.040	[-8.37, 5.22]	0.954	-

Interpretation. $\text{Gap}_R = 0.015$ [0.006, 0.031]: English refusal outcomes predicted Slovene refusal almost perfectly ($R2^* = 0.968$), with only 1.5% of the ceiling left unexplained. This gap, while statistically above zero, is small. The blind share B_t for R was 0.004 with a CI spanning zero, meaning the small gap is mostly noise rather than a systematic edit-driven asymmetry. Verdict: CONFIRMATORY. Refusal is visible from English for GaMS3.

$\text{Gap}_K = 0.147$ [0.074, 0.265]: KL divergence showed a substantial surrogacy gap. English KL predicted Slovene KL at $R2^* = 0.830$, leaving 14.7% of the ceiling unexplained. The blind share

$B_t = 0.050$ [0.001, 0.121], with the CI excluding zero: part of this gap is systematically determined by the edit, not just noise. Verdict: BLIND. KL divergence has a component in Slovene that English KL does not see. Holm $p = 0.000$.

Gap_N and Gap_M showed no reliable signal (NLL and max-logit traits had low reliability in English, so the gap estimate was uninformative).

Carrier regression. The exposure differential D ($\log E_{SL} - \log E_{EN}$) was tested as a carrier of the surrogacy gap. For every trait, $dR2(D \text{ given } S0)$ was near zero or negative. The D carrier FAILED for all traits. The surrogacy gap is not explained by differences in how much the edit perturbs Slovene vs. English token representations on harmless text.

Transfer slopes. The SL/EN transfer slope for R was 0.82 [0.74, 0.90]: when the edit moves English refusal by one unit, Slovene refusal moves by 0.82 units. For K, the slope was 0.39 [0.34, 0.48], much stronger attenuation. For R1, the slope was 0.50 [0.46, 0.55].

Forecast. Conformal prediction intervals at 90% coverage met the 85% threshold on only 50% of trait $\times$ holdout-set combinations. The core edit (trial 88) was out of support for all traits, meaning it lay outside the distribution of TPE-explored edits. This limits the forecast's applicability to the specific selected checkpoint.

Stability checks. Gap estimates were stable across alternative learners, inclusion of collapsed edits, reverse-direction analysis (SL $\rightarrow$ EN), and MT-stable subsets.

Experiment 7: Gemma surrogacy panel

Goal. Measure the English surrogacy gap for Gemma using the same protocol as experiment 6.

Setup. 203 Heretic edits, 163 fitted. The same six traits were measured.

Verdicts:

- **P-a (Gap_R is small, ≤ 0.05):** NOT CONFIRMED. $\text{Gap}_{R1} = 0.054$ [0.022, 0.128]. The gap exceeded the pre-registered threshold of 0.05 for R1.
- **P-b (Gap_Rb is small):** UNTESTABLE. The Rb validity gate failed, so the analysis was exploratory. Exploratory estimate: $\text{Gap}_{Rb} = 0.018$ [0.011, 0.043].
- **P-c (exposure differential carries the gap):** NOT CONFIRMED. $dR2(P \text{ given base})$ and $dR2(b1 \text{ given } P)$ were near zero for both R1 and Rb.
- **P-d (matched-margin gap is small):** NOT CONFIRMED. $\text{Gap}_{\text{margin_matched}} = 0.057$ [0.029, 0.130].

Key result. $\text{Gap}_K = +0.079$ (Holm $p = 0.009$): KL divergence showed a surrogacy gap for Gemma as well, though smaller than GaMS3's 0.147. The SL/EN transfer slope for R1 was 0.436, indicating strong attenuation: when the edit moved English first-token refusal by one unit, Slovene moved only 0.44 units.

Comparison with GaMS3. The Gemma surrogacy gap on refusal traits was larger than GaMS3's (0.054 vs. 0.015 for R1), consistent with the behavioural result that Gemma retained more Slovene refusal after editing. The KL gap was smaller (0.079 vs. 0.147), but both models showed a significant blind component in KL.

Experiment 8: Causal test

Goal. Test whether the Slovene refusal that survived English ablation can be causally removed by specific interventions, and identify where in the network the relevant information resides.

Pre-registered hypothesis. Subtracting the false-refusal direction r_prior (constructed from prompts judged harmless that the model refuses, following Wang et al. [2]) from the English refusal direction d_EN should cut the residual Slovene refusal by at least half, beyond what a random control achieves. The KILL criterion fires if the r_prior cut is not significantly larger than the best random control.

Primary result: hypothesis FAILS. Applying d_EN alone reduced Gemma's harmful EN refusal from 0.91 to 0.23 but left SL refusal at 0.86. Adding r_prior (arm A2: $d_EN + r_prior$) moved SL refusal to 0.83. The r_prior cut relative to d_EN alone was 0.036 [-0.037, 0.109], not significantly different from zero, and not significantly larger than the best random control (0.074). KILL criteria a and c fired (part 1 failed, and F6 over-refusal test failed). The pre-registered hypothesis was rejected.

Full Gemma activation arms (111 harmful + 111 harmless per language):

Arm	EN harmful	SL harmful	EN harmless	SL harmless
A0 no-op	0.91	0.99	0.05	0.28
A1 d_EN	0.23	0.86	0.00	0.19
A2 $d_EN + r_prior$	0.21	0.83	0.00	0.12
A3 r_prior alone	0.95	0.98	0.04	0.19
A4 $d_EN +$ language identity	0.23	0.55	0.02	0.08
A7 $d_EN + rand_1$	0.21	0.85	0.01	0.15
A10 $d_EN +$ shuffled-label r_prior	0.41	0.73	-	-

Several observations from the arm table:

1. **r_prior alone (A3)** had almost no effect: SL harmful refusal went from 0.99 to 0.98, EN from 0.91 to 0.95. The false-refusal direction does not carry meaningful refusal-suppression information by itself.
1. **Language-identity direction (A4)** combined with d_EN reduced SL harmful refusal from 0.86 to 0.55, the largest single-direction improvement. However, it also raised FLORES NLL by 2.0 (EN) and 1.9 (SL), destroying language modelling quality. This is not a practical intervention.
1. **Random directions (A7, A8, A9)** combined with d_EN left SL refusal between 0.74 and 0.85, comparable to d_EN alone. Random directions did not systematically help.
1. **Shuffled-label r_prior (A10)** combined with d_EN reduced SL refusal to 0.73 but also raised EN harmful refusal to 0.41, suggesting the shuffled direction disrupted the English refusal pathway as well.

Exploratory: layer-matched d_EN across all 48 layers (X1). Applying the per-layer English

refusal direction at all 48 layers simultaneously reduced SL harmful refusal from 0.86 to 0.10 and EN harmful refusal to 0.03. This is the only intervention that nearly eliminated Slovene refusal, but it required steering at every layer simultaneously, raised FLORES NLL by 0.29 (EN) and 0.52 (SL), and is not equivalent to the single-direction ablation that Heretic performs.

Depth localisation. Applying layer-matched d_EN to different layer bands:

Band	SL harmful	EN harmful	FLORES Δ NLL SL
Layers 1-12	0.82	0.79	0.743
Layers 13-24	0.82	0.45	0.017
Layers 25-36	0.70	0.72	0.082
Layers 37-48	0.99	0.87	-0.003
Layers 1-24	0.55	0.42	0.409
Layers 1-36	0.21	0.38	0.513
All 48	0.23	0.37	0.524

No single band removed Slovene refusal. Layers 13-24 were most effective for English (0.45) but left Slovene at 0.82. Layers 25-36 provided the most Slovene reduction per band (0.70) without large NLL cost. Full Slovene refusal removal required cumulative intervention across at least three bands.

Weight edit repair. The experiment also tested whether augmenting the core Heretic weight edit with additional directions could repair the Slovene gap:

Arm	SL harmful	EN harmful	FLORES Δ NLL SL
W0 core Heretic edit	0.89	0.39	-0.005
W1 core + r_prior weight edit	0.76	0.23	0.009
W3 core + layer-matched d_EN(h)	0.11	0.29	0.546
W4 core + random weight edit	0.47	0.70	0.064

The r_prior weight edit (W1) reduced SL refusal from 0.89 to 0.76 while also reducing EN refusal to 0.23, but the SL residual remained high. The layer-matched weight edit (W3) reached 0.11 SL refusal but at severe NLL cost (+0.55). The random weight edit (W4) reduced SL refusal to 0.47 while raising EN refusal to 0.70, suggesting it disrupted the model rather than targeting Slovene refusal.

Community Heretic reference vs. core edit (S4 held-out, 70 pairs):

Arm	EN harmful	SL harmful	SL harmless
W0 core edit	0.39	0.89	0.07
C0 community edit	0.01	0.26	0.01
C1 community + r_prior	0.01	0.20	0.00

The community edit reached much lower SL refusal (0.26) than the selected trial's edit (0.89), at the cost of slightly higher FLORES NLL (+0.11 SL). Adding r_prior to the community edit reduced SL refusal from 0.26 to 0.20, a modest improvement. The community edit achieved stronger refusal suppression because it was more aggressive overall, not because it targeted Slovene specifically.

GaMS3 descriptive arms. The GaMS3 results were descriptive (not pre-registered for the causal test):

Arm	EN harmful	SL harmful	EN harmless	SL harmless
G0 no-op	0.93	0.95	0.07	0.10
G1 d_EN	0.57	0.50	0.00	0.04

For GaMS3, applying d_EN reduced both EN and SL refusal comparably (from 0.93/0.95 to 0.57/0.50), consistent with the small surrogacy gap found in experiment 6. The EN-SL difference was minor.

Summary of findings

[Figure fig.refusal_gap: generated in the report phase, not in this draft]

The central result across both iterations is a dissociation between GaMS3 and Gemma in how English-tuned ablation transfers to Slovene.

GaMS3 showed near-complete transfer. The English-tuned Heretic edit reduced refusal to $\leq 6\%$ in both languages (experiment 5, T1). The surrogacy gap on refusal was 0.015, meaning English outcomes predicted Slovene outcomes almost perfectly (experiment 6). The frozen probe collapsed after editing (experiment 5, T6), indicating the edit destroyed the harm signal at the representation level. Because GaMS3 had continual Slovene pretraining on top of the same Gemma-3 base, its harm representation may have become more tightly integrated across languages, allowing a single English direction to capture both.

Gemma showed a large transfer gap. The edit reduced English refusal from 0.971 to 0.287 on RefusEU but left Slovene refusal at 0.739 (experiment 4). The surrogacy gap on first-token refusal was 0.054 (experiment 7). The frozen probe was preserved after editing (experiment 5, T6), meaning the harm signal remained intact in the representation: the edit changed the readout for English but not for Slovene. The causal test (experiment 8) found that no single direction combined with d_EN eliminated the Slovene residual; only cumulative steering across all 48 layers achieved near-zero Slovene refusal, at substantial NLL cost.

The exposure differential D did not carry the gap in either model (experiments 6 and 7). The surrogacy gap is not explained by differences in how much the edit perturbs Slovene vs. English representations on harmless text.

KL divergence showed a blind surrogacy gap in both models: Gap_K = 0.147 for GaMS3 and 0.079 for Gemma. The English KL constraint in Heretic's objective does not fully capture what the edit does to Slovene output distributions.

Utility was preserved in all cells: macro task changes ≤ 0.20 points, FLORES NLL changes ≤ 0.002 (experiments 5, T3-T4). The edits did not damage general capabilities.

Over-refusal decreased after editing in both models (experiment 4). GaMS3 edit reduced S6 over-refusal from $\sim 10\%$ to 0%. Gemma edit reduced EN over-refusal from 8.7% to 1.3% and SL over-refusal from 32.7% to 19.3%. The original Gemma model had high Slovene over-refusal, and the edit only partially addressed it.

Dead ends and null results. (1) The A3 screen (experiment 3) for GaMS3 failed its validity gate, so no quantitative screen verdict was obtainable for that model. (2) The r_prior false-refusal

direction did not carry meaningful Slovene residual in the causal test. (3) The exposure differential D, hypothesised to explain the surrogacy gap, showed zero or negative dR2 across all traits and both models. (4) Conformal prediction intervals achieved adequate coverage on only half of trait $\times$ holdout-set combinations, limiting the forecasting value of the surrogacy panel. (5) Cross-machine reproducibility checks showed non-bit-identical results (max |diff| up to 97.2), attributable to NF4 quantisation non-determinism.

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